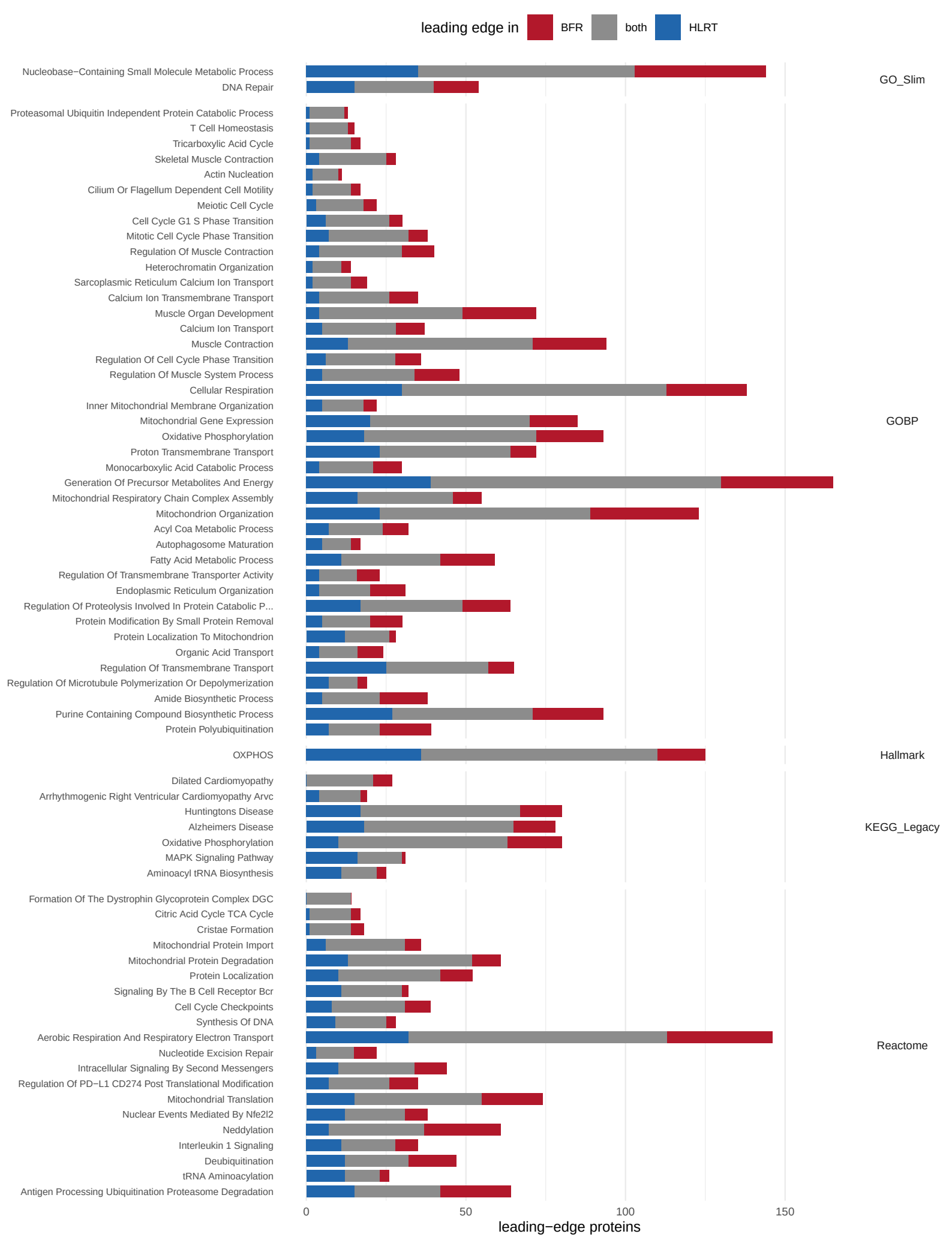
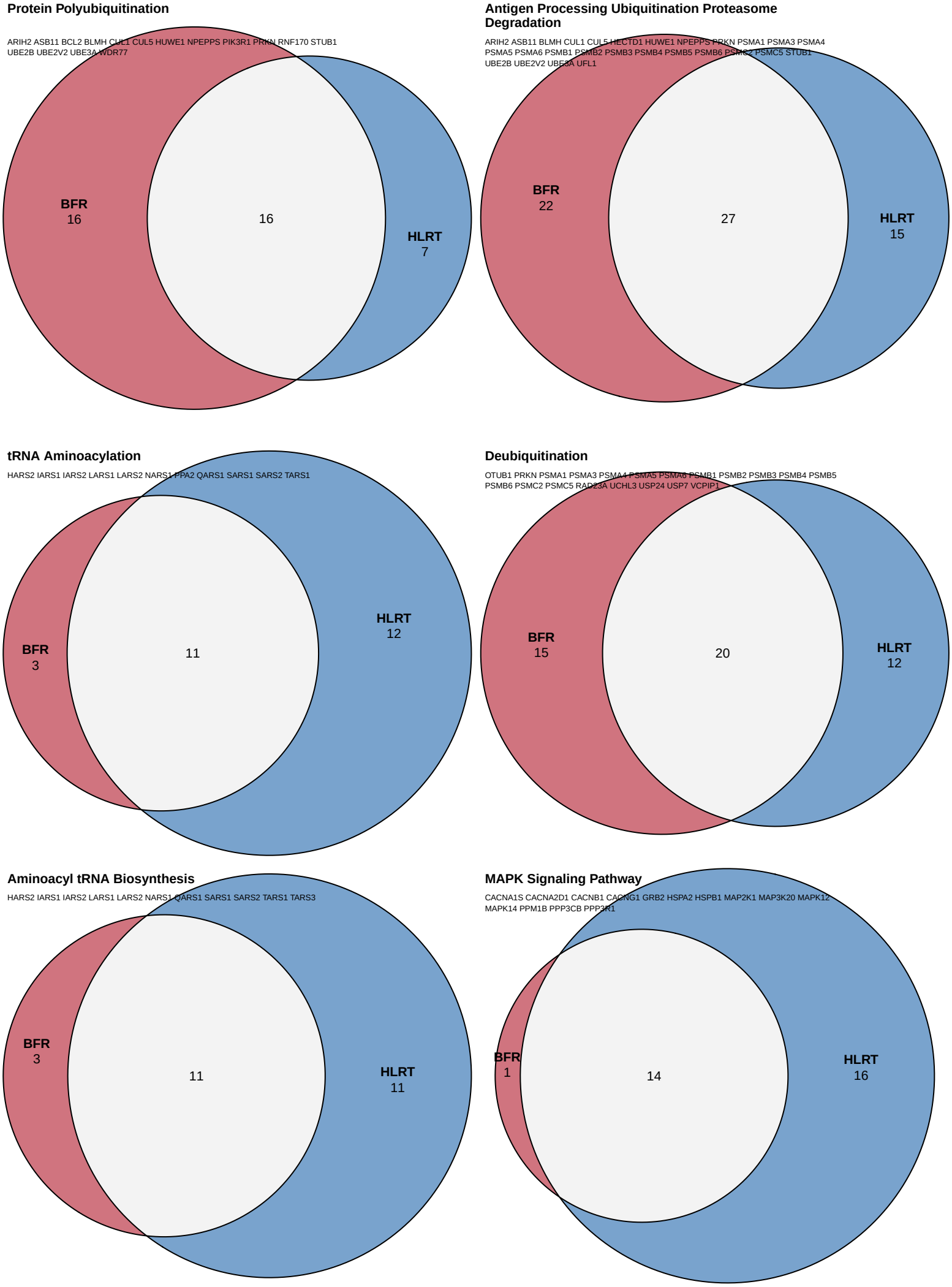
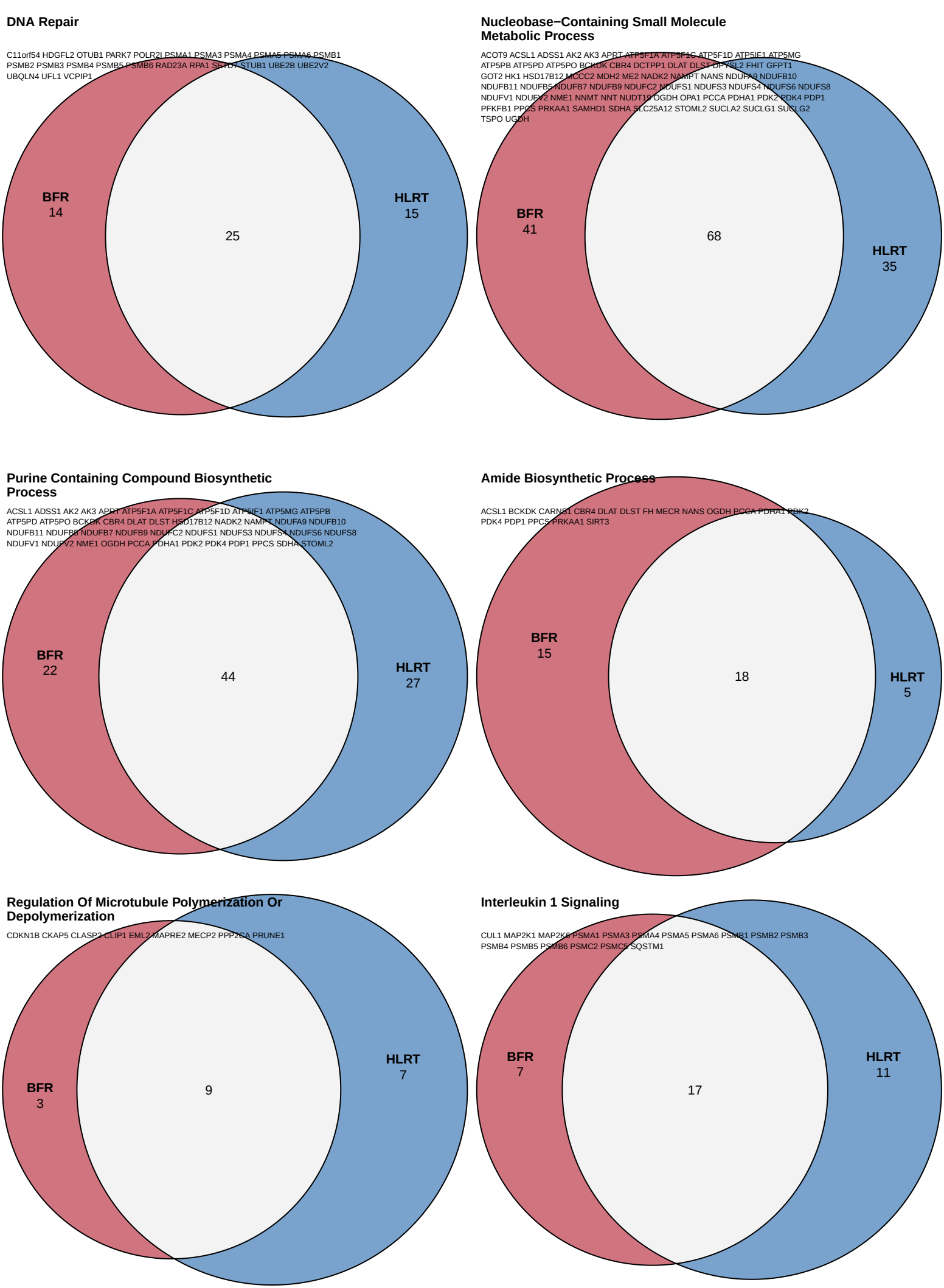


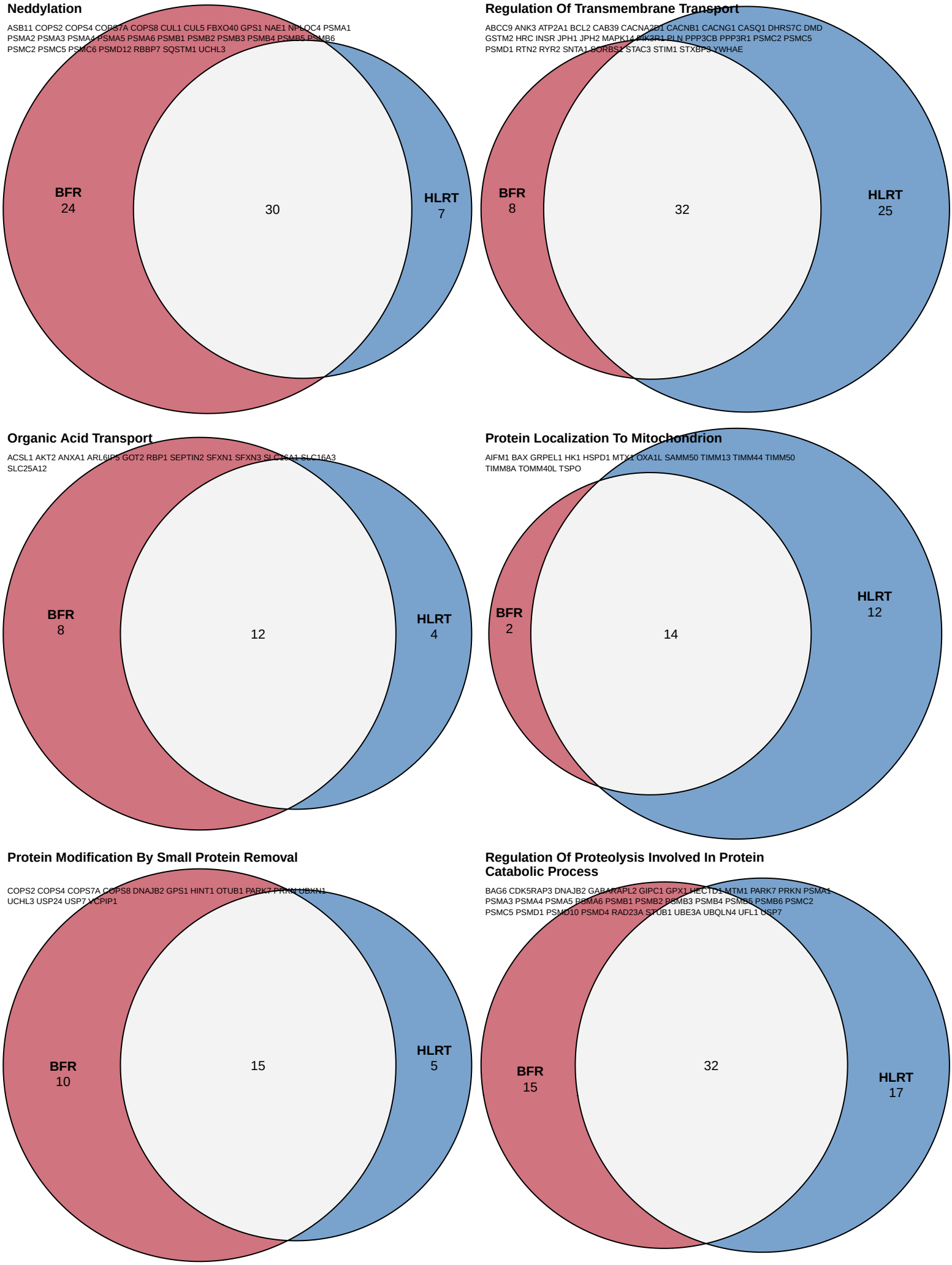
S20 Figure. Leading−edge overlap between the legs. Every pathway that fgsea calls at FDR < 0.05 in both training contrasts, in the same direction (71, each a main set after collapse in at least one leg). Bars split each pathway's leading−edge proteins into those shared by both legs and those in one leg only; lowest overlap at the bottom of each collection. Leading edges move with small ranking changes, so this describes and does not test. Data: leading_edge sheet of 07_leading_edge.xlsx.



S21 Figure (page 1 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



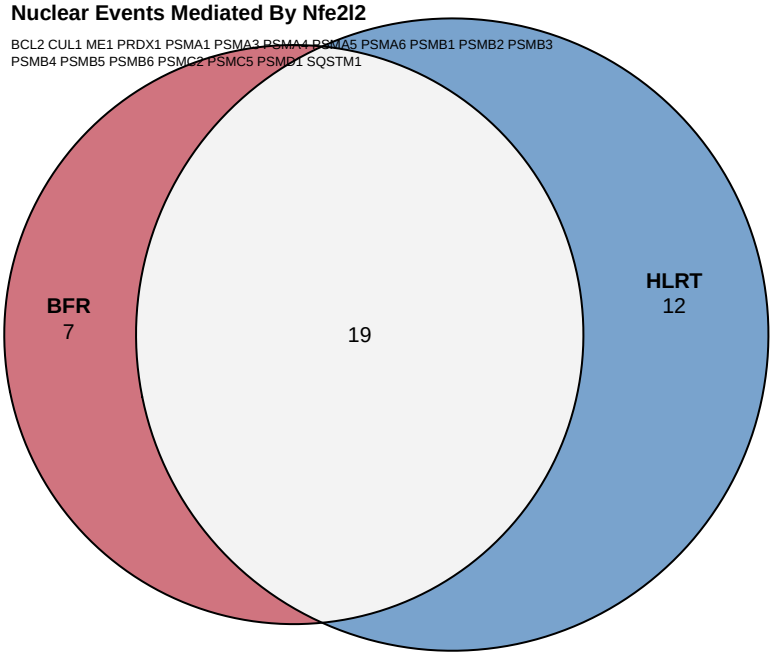
S21 Figure (page 2 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



S21 Figure (page 3 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.

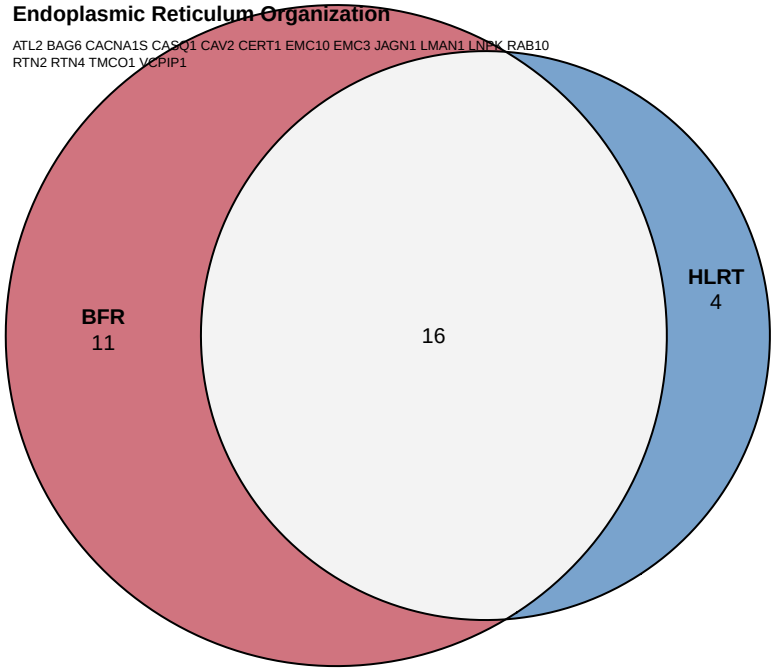
Nuclear Events Mediated By Nfe2l2

BCL2 CUL1 ME1 PRDX1 PSMA1 PSMA3 PSMA4 PSMA5 PSMA6 PSMB1 PSMB2 PSMB3
PSMB4 PSMB5 PSMB6 PSMC2 PSMC5 PSMD1 SQSTM1



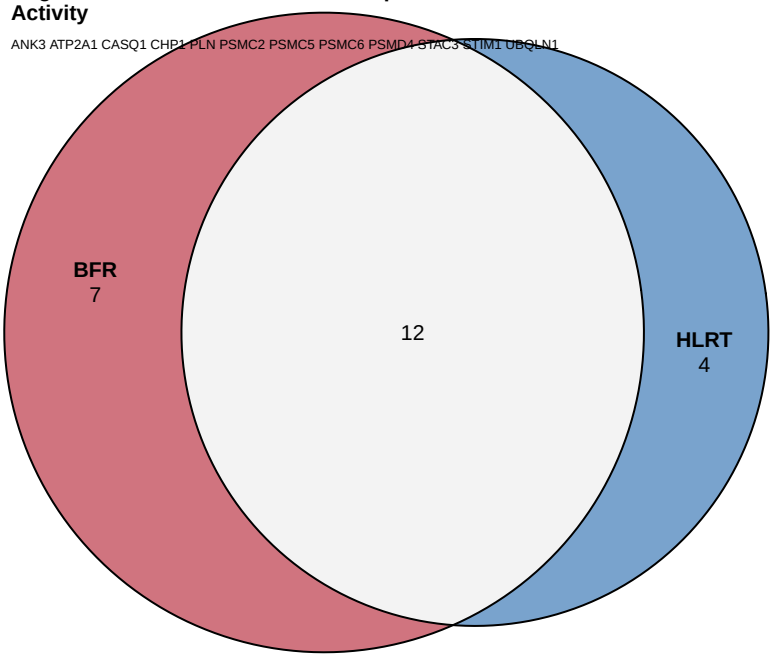
Endoplasmic Reticulum Organization

ATL2 BAG6 CACNA1S CASQ1 CAV2 CERT1 EMC10 EMC3 JAGN1 LMAN1 LNPK RAB10
RTN2 RTN4 TMCO1 VCPIP1



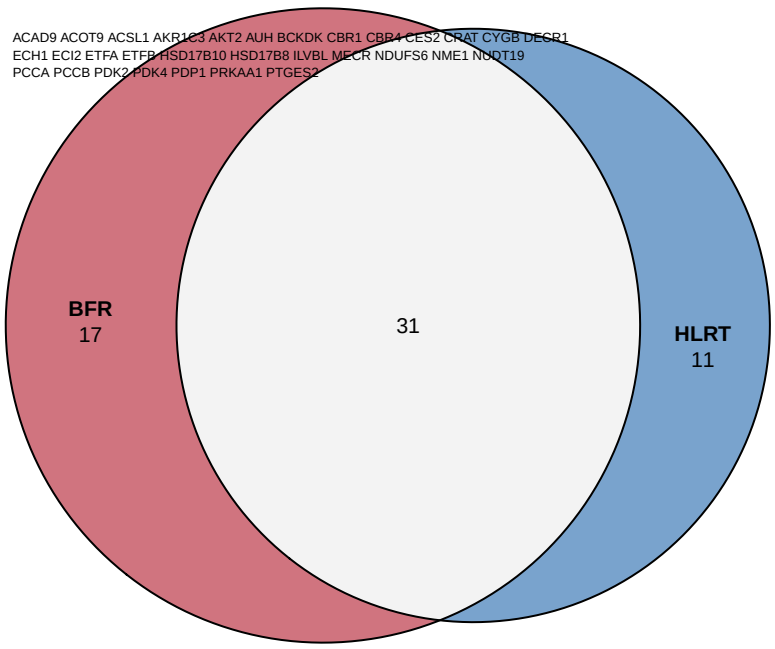
Regulation Of Transmembrane Transporter Activity

ANK3 ATP2A1 CASQ1 CHP1 PLN PSMC2 PSMC5 PSMC6 PSMD4 STAC3 STIM1 UBQLN1



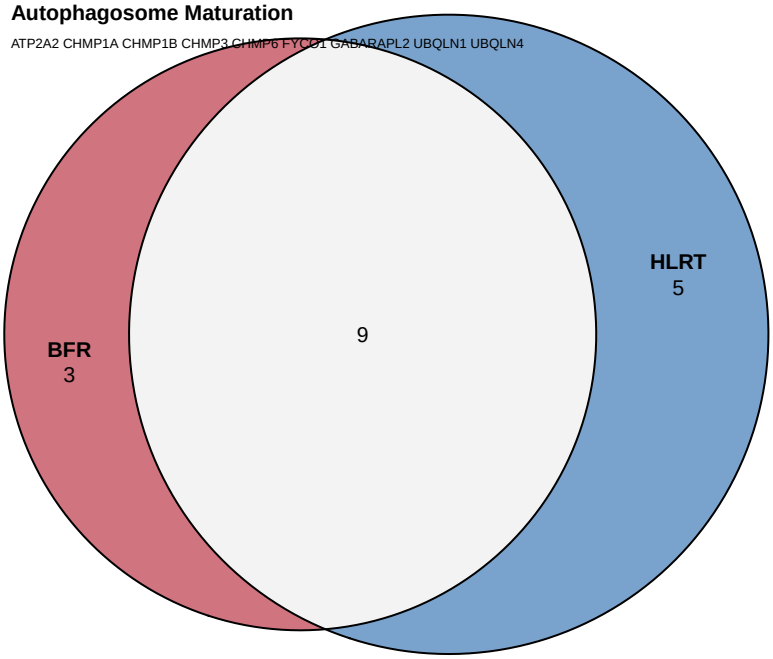
Fatty Acid Metabolic Process

ACAD9 ACOT9 ACSL1 AKR1C3 AKT2 AUH BCKDK CBR1 CBR4 CES2 CRAT CYGB DECR1
ECH1 ECI2 ETFA ETFB HSD17B10 HSD17B8 ILVBL MECR NDUFS6 NME1 NUDT19
PCCA PCCB PDK2 PDK4 PDP1 PRKAA1 PTGES2



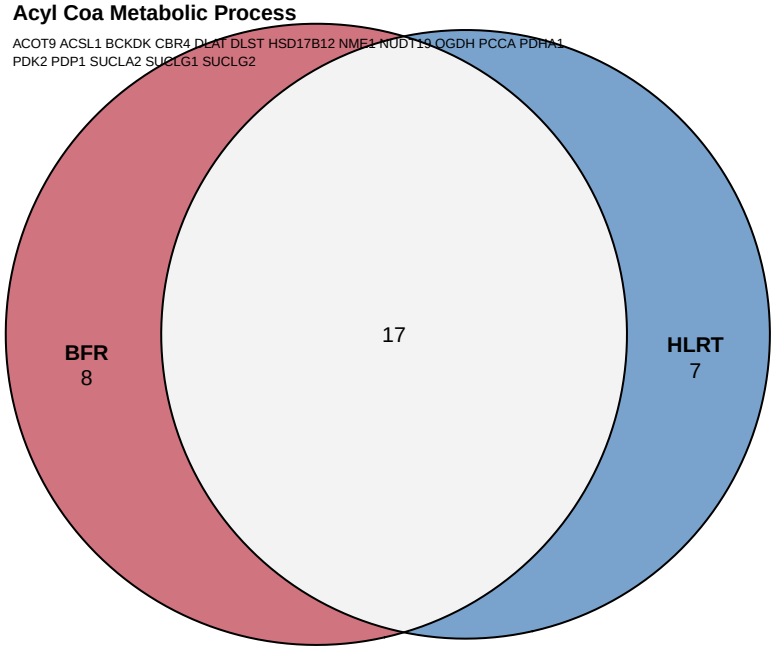
Autophagosome Maturation

ATP2A2 CHMP1A CHMP1B CHMP3 CHMP6 FYCO1 GABARAPL2 UBQLN1 UBQLN4



Acyl Coa Metabolic Process

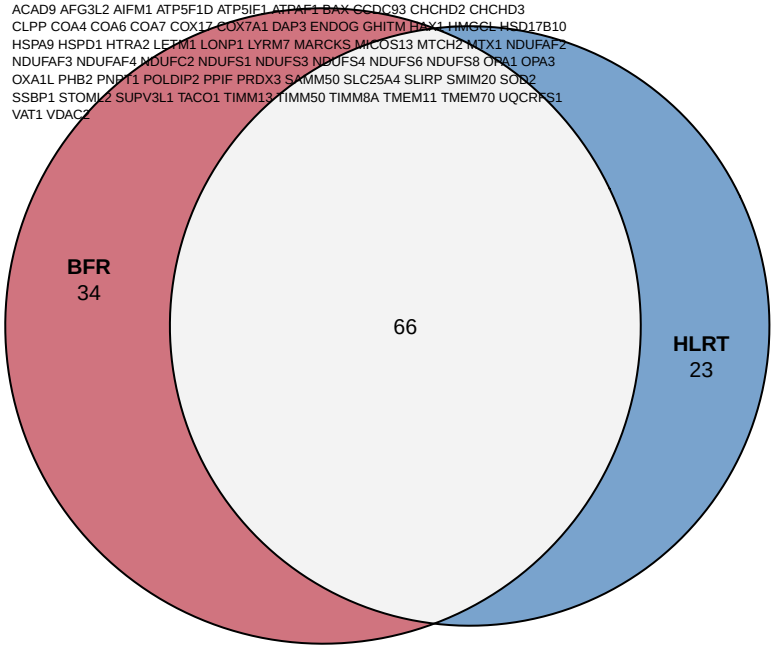
ACOT9 ACSL1 BCKDK CBR4 DLAT DLST HSD17B12 NME1 NUDT19 OGDH PCCA PDHA1
PDK2 PDP1 SUCLA2 SUCLG1 SUCLG2



S21 Figure (page 4 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.

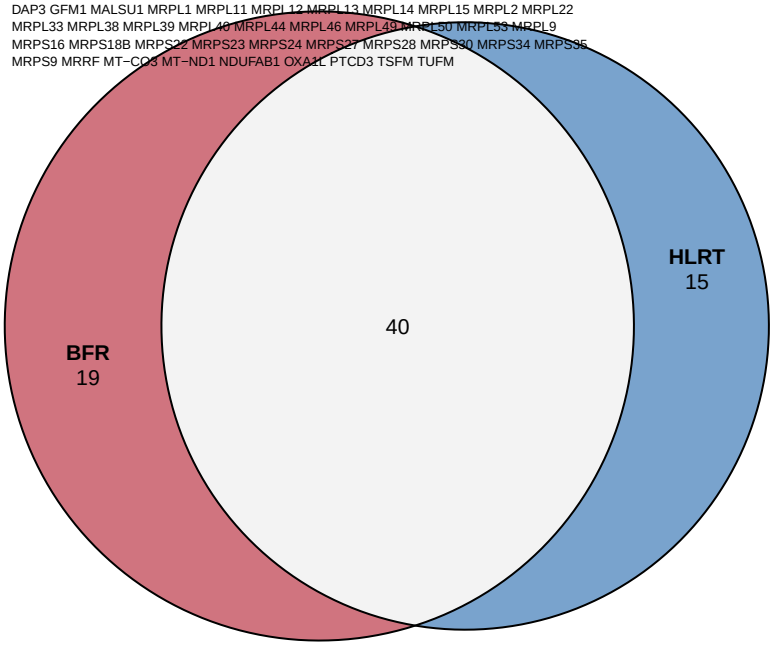
Mitochondrion Organization

ACAD9 AFG3L2 AIFM1 ATP5F1D ATP5IF1 ATPAF1 BAX CCDC93 CHCHD2 CHCHD3
CLPP COA4 COA6 COA7 COX17 COX7A1 DAP3 ENDOG GHITM HAX1 HMGCL HSD17B10
HSPA9 HSPD1 HTRA2 LETM1 LONP1 LYRM7 MARCKS MICOS13 MTCH2 MTX1 NDUFAF2
NDUFAF3 NDUFAF4 NDUFC2 NDUFS1 NDUFS3 NDUFS4 NDUFS6 NDUFS8 OPA1 OPA3
OXA1L PHB2 PNPT1 POLDIP2 PPIF PRDX3 SAMM50 SLC25A4 SLIRP SMIM20 SOD2
SSBP1 STOML2 SUPV3L1 TACO1 TIMM13 TIMM50 TIMM8A TMEM11 TMEM70 UQCRFS1
VAT1 VDAC2



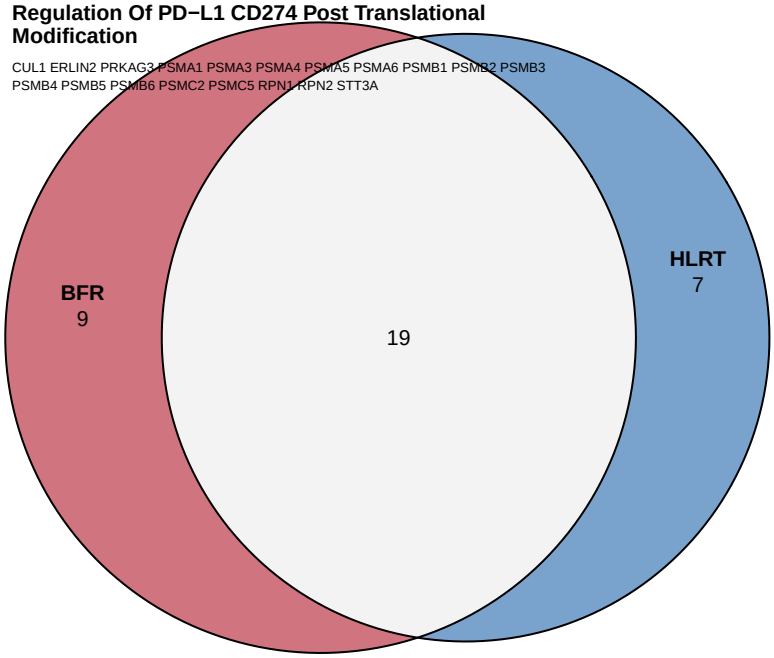
Mitochondrial Translation

DAP3 GFM1 MALSU1 MRPL1 MRPL11 MRPL12 MRPL13 MRPL14 MRPL15 MRPL2 MRPL22
MRPL33 MRPL38 MRPL39 MRPL40 MRPL44 MRPL46 MRPL49 MRPL50 MRPL53 MRPL9
MRPS16 MRPS18B MRPS22 MRPS23 MRPS24 MRPS27 MRPS28 MRPS30 MRPS34 MRPS35
MRPS9 MRRF MT-CO3 MT-ND1 NDUFAB1 OXA1L PTCD3 TSFM TUFM



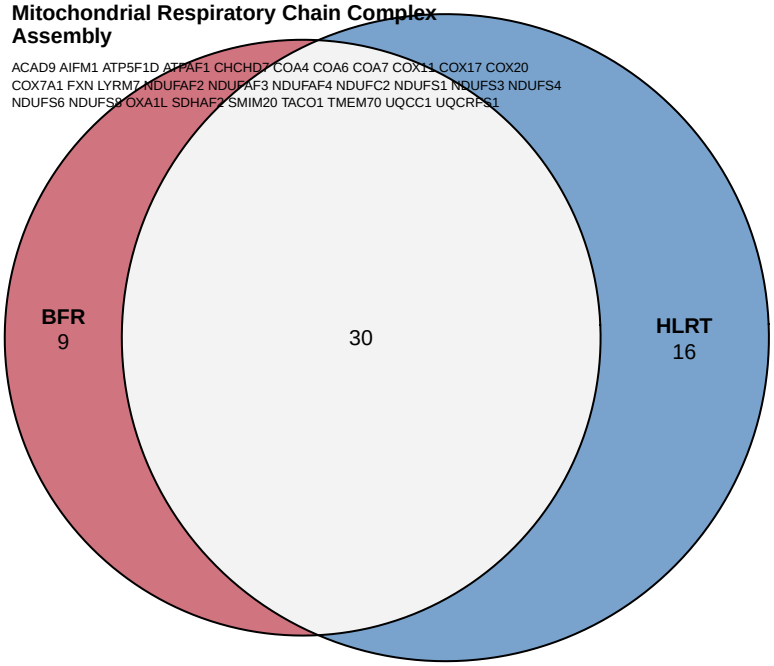
Regulation Of PD-L1 CD274 Post Translational Modification

CUL1 ERLIN2 PRKAG3 PSMA1 PSMA3 PSMA4 PSMA5 PSMA6 PSMB1 PSMB2 PSMB3
PSMB4 PSMB5 PSMB6 PSMC2 PSMC5 RPN1 RPN2 STT3A



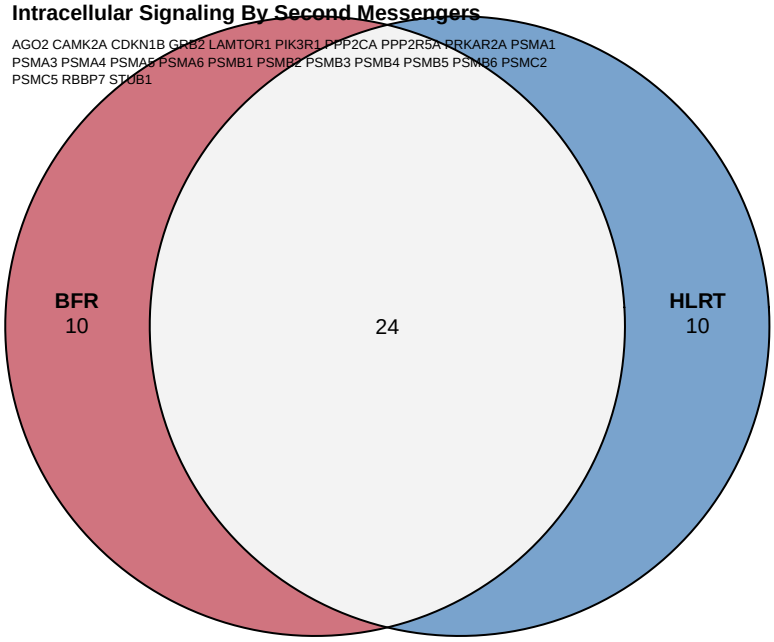
Mitochondrial Respiratory Chain Complex Assembly

ACAD9 AIFM1 ATP5F1D ATPAF1 CHCHD7 COA4 COA6 COA7 COX11 COX17 COX20
COX7A1 FXN LYRM7 NDUFAF2 NDUFAF3 NDUFAF4 NDUFC2 NDUFS1 NDUFS3 NDUFS4
NDUFS6 NDUFS8 OXA1L SDHAF2 SMIM20 TACO1 TMEM70 UQCC1 UQCRFS1



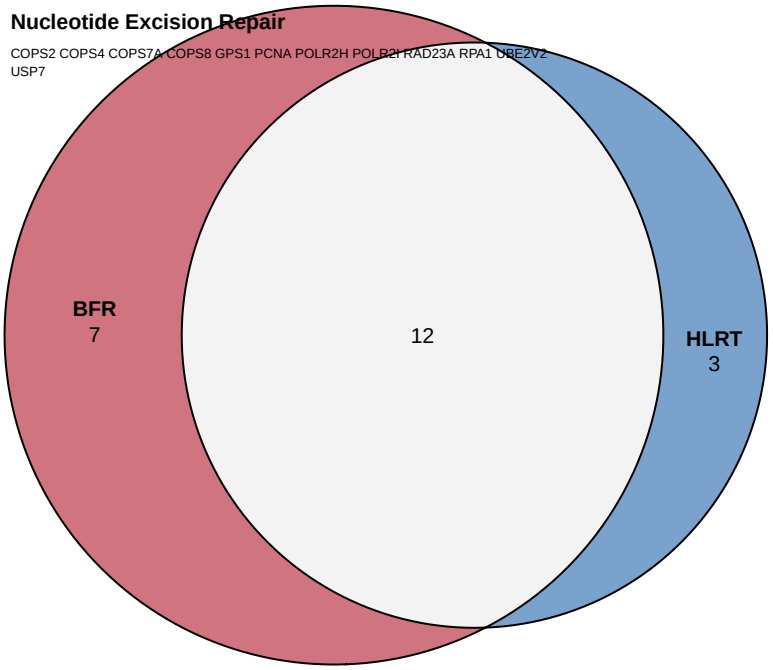
Intracellular Signaling By Second Messengers

AGO2 CAMK2A CDKN1B GRB2 LAMTOR1 PIK3R1 PPP2CA PPP2R5A PRKAR2A PSMA1
PSMA3 PSMA4 PSMA5 PSMA6 PSMB1 PSMB2 PSMB3 PSMB4 PSMB5 PSMB6 PSMC2
PSMC5 RBBP7 STUB1



Nucleotide Excision Repair

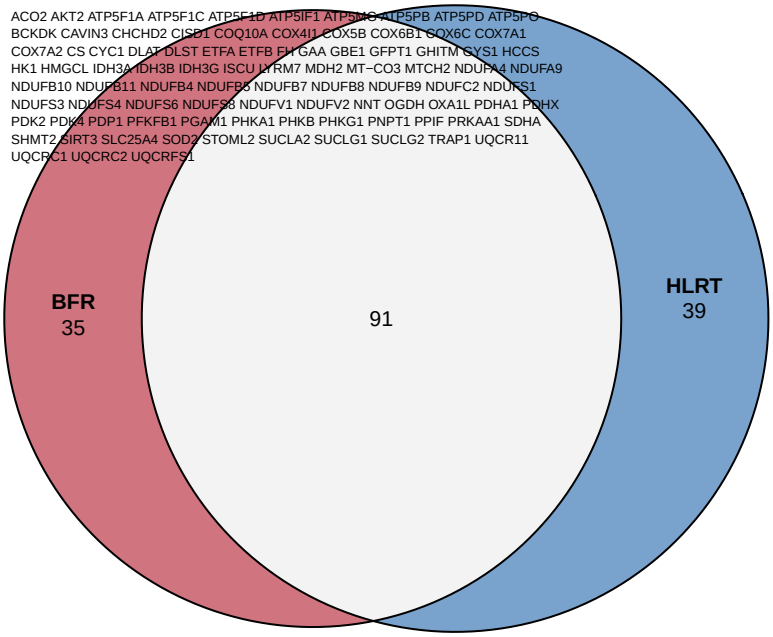
COPS2 COPS4 COPS7A COPS8 GPS1 PCNA POLR2H POLR2I RAD23A RPA1 UBE2V2
USP7



S21 Figure (page 5 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.

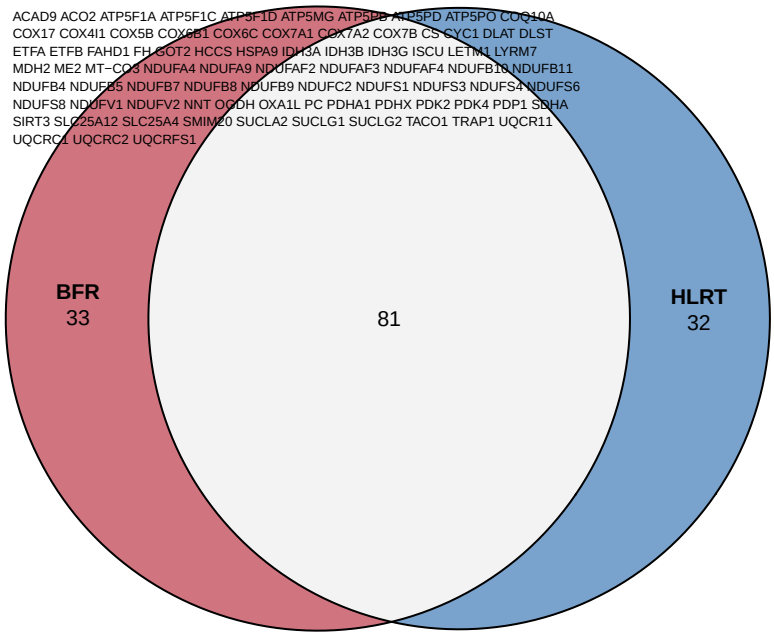
Generation Of Precursor Metabolites And Energy

ACO2 AKT2 ATP5F1A ATP5F1C ATP5F1D ATP5IF1 ATP5MG ATP5PB ATP5PD ATP5PO BCKDK CAVIN3 CHCHD2 CISD1 COQ10A COX4I1 COX5B COX6B1 COX6C COX7A1 COX7A2 CS CYC1 DLAT DLST ETFA ETFB FH GAA GBE1 GFPT1 GHITM GYS1 HCCS HK1 HMGCL IDH3A IDH3B IDH3G ISCU LYRM7 MDH2 MT−CO3 MTCH2 NDUFA4 NDUFA9 NDUFB10 NDUFB11 NDUFB4 NDUFB5 NDUFB7 NDUFB8 NDUFB9 NDUFC2 NDUFS1 NDUFS3 NDUFS4 NDUFS6 NDUFS8 NDUFV1 NDUFV2 NNT OGDH OXA1L PDHA1 PDHX PDK2 PDK4 PDP1 PFKFB1 PGAM1 PHKA1 PHKB PHKG1 PNPT1 PPIF PRKAA1 SDHA SHMT2 SIRT3 SLC25A4 SOD2 STOML2 SUCLA2 SUCLG1 SUCLG2 TRAP1 UQCR11 UQCRC1 UQCRC2 UQCRFS1



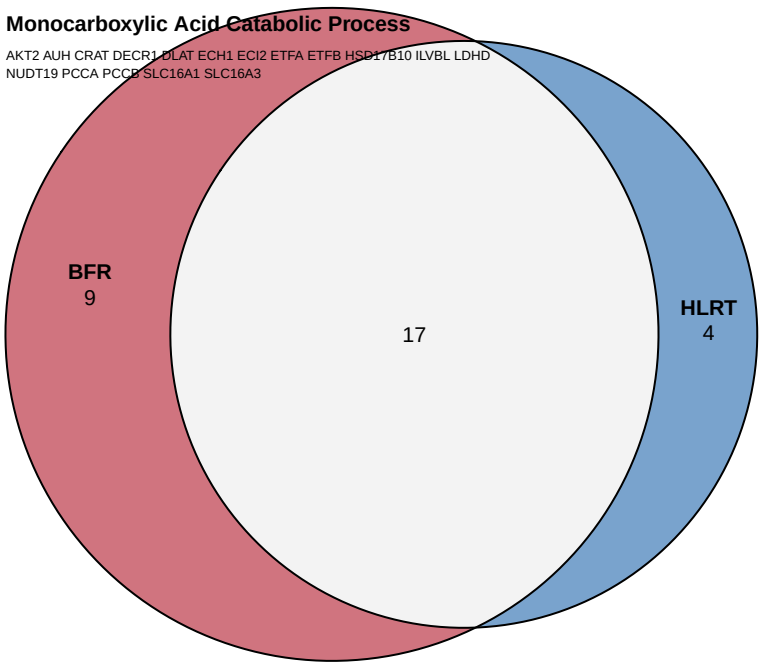
Aerobic Respiration And Respiratory Electron Transport

ACAD9 ACO2 ATP5F1A ATP5F1C ATP5F1D ATP5MG ATP5PB ATP5PD ATP5PO COQ10A COX17 COX4I1 COX5B COX6B1 COX6C COX7A1 COX7A2 COX7B CS CYC1 DLAT DLST ETFA ETFB FAHD1 FH GOT2 HCCS HSPA9 IDH3A IDH3B IDH3G ISCU LETM1 LYRM7 MDH2 ME2 MT−CO3 NDUFA4 NDUFA9 NDUFAF2 NDUFAF3 NDUFAF4 NDUFB10 NDUFB11 NDUFB4 NDUFB5 NDUFB7 NDUFB8 NDUFB9 NDUFC2 NDUFS1 NDUFS3 NDUFS4 NDUFS6 NDUFS8 NDUFV1 NDUFV2 NNT OGDH OXA1L PC PDHA1 PDHX PDK2 PDK4 PDP1 SDHA SIRT3 SLC25A12 SLC25A4 SMIM20 SUCLA2 SUCLG1 SUCLG2 TACO1 TRAP1 UQCR11 UQCRC1 UQCRC2 UQCRFS1



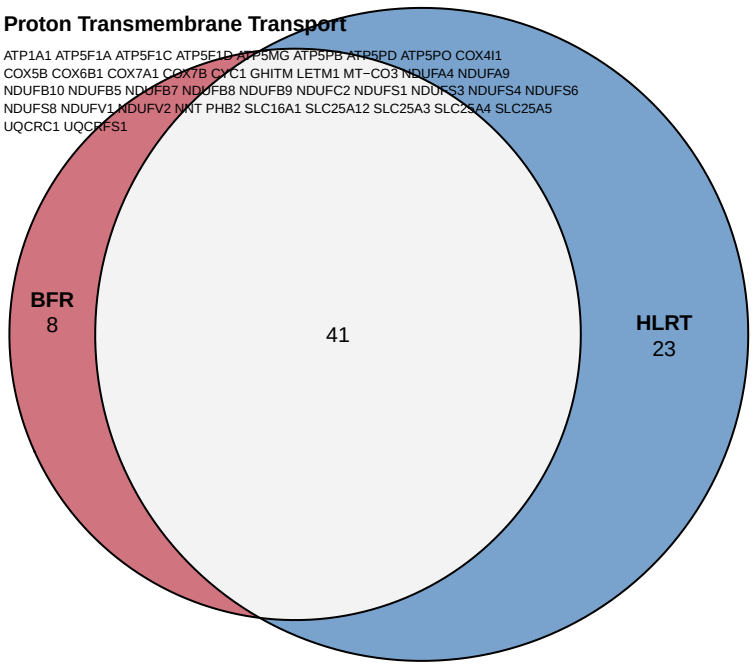
Monocarboxylic Acid Catabolic Process

AKT2 AUH CRAT DECR1 DLAT ECH1 ECI2 ETFA ETFB HSD17B10 ILVBL LDHD NUDT19 PCCA PCCB SLC16A1 SLC16A3



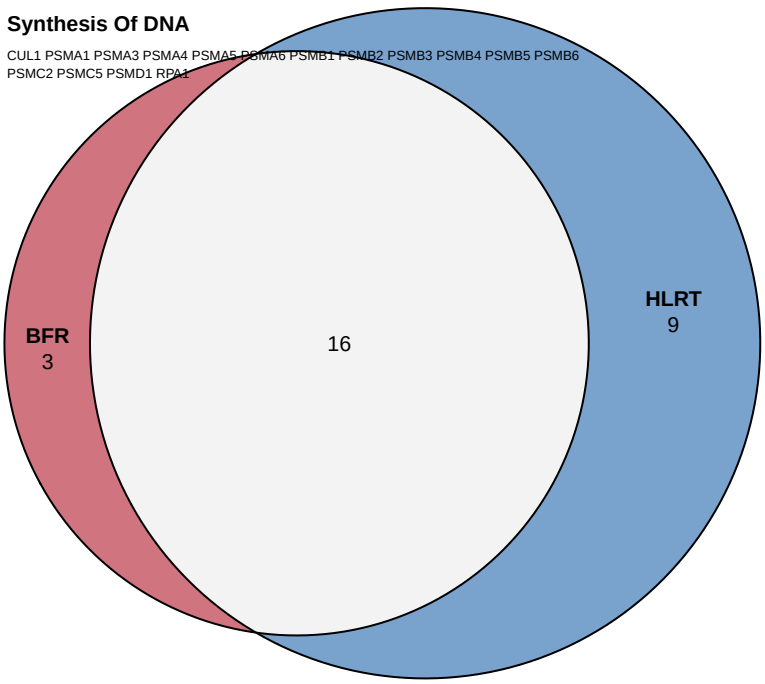
Proton Transmembrane Transport

ATP1A1 ATP5F1A ATP5F1C ATP5F1D ATP5MG ATP5PB ATP5PD ATP5PO COX4I1 COX5B COX6B1 COX7A1 COX7B CYC1 GHITM LETM1 MT−CO3 NDUFA4 NDUFA9 NDUFB10 NDUFB5 NDUFB7 NDUFB8 NDUFB9 NDUFC2 NDUFS1 NDUFS3 NDUFS4 NDUFS6 NDUFS8 NDUFV1 NDUFV2 NNT PHB2 SLC16A1 SLC25A12 SLC25A3 SLC25A4 SLC25A5 UQCRC1 UQCRFS1



Synthesis Of DNA

CUL1 PSMA1 PSMA3 PSMA4 PSMA5 PSMA6 PSMB1 PSMB2 PSMB3 PSMB4 PSMB5 PSMB6 PSMC2 PSMC5 PSMD1 RPA1

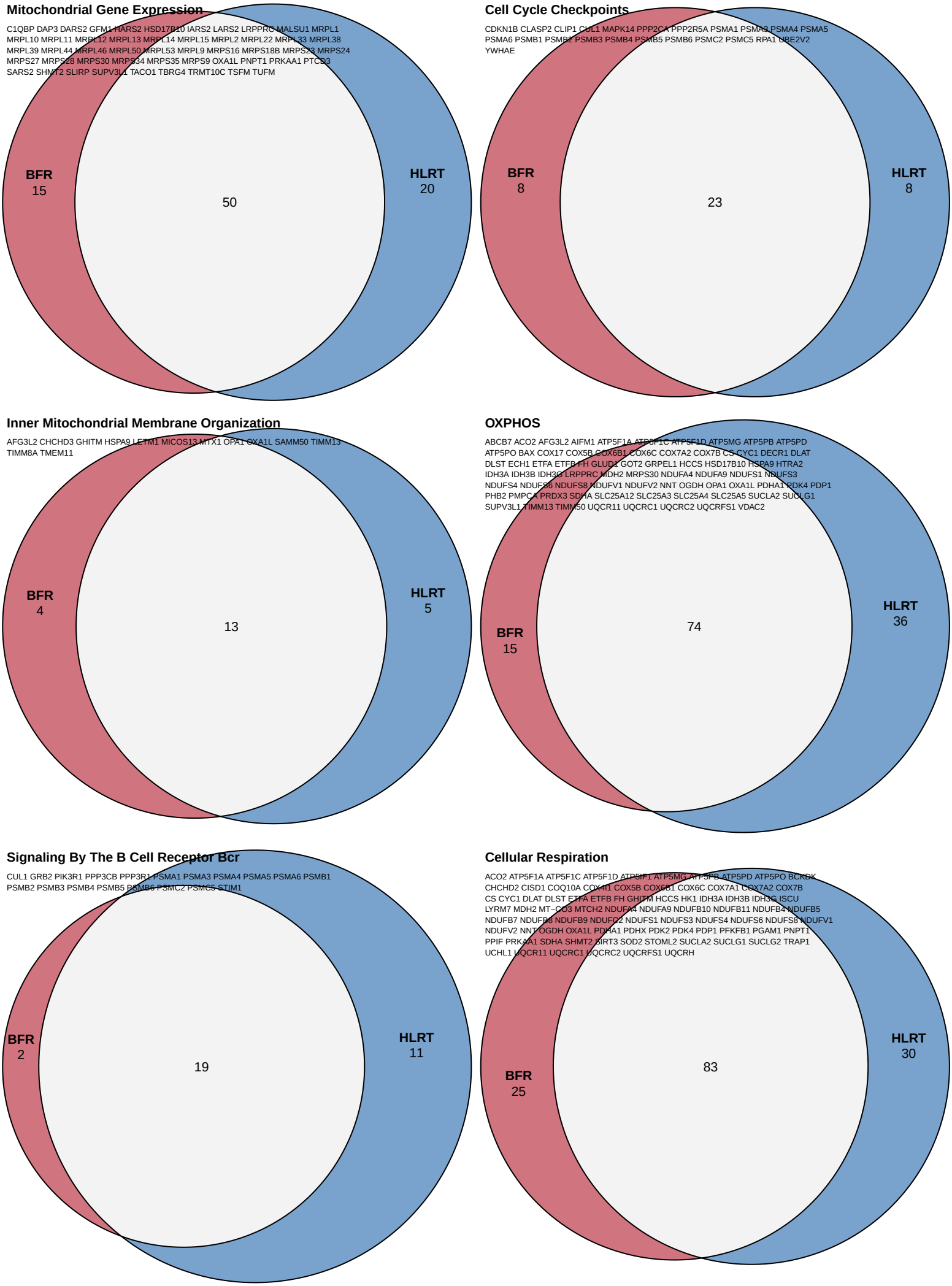


Oxidative Phosphorylation

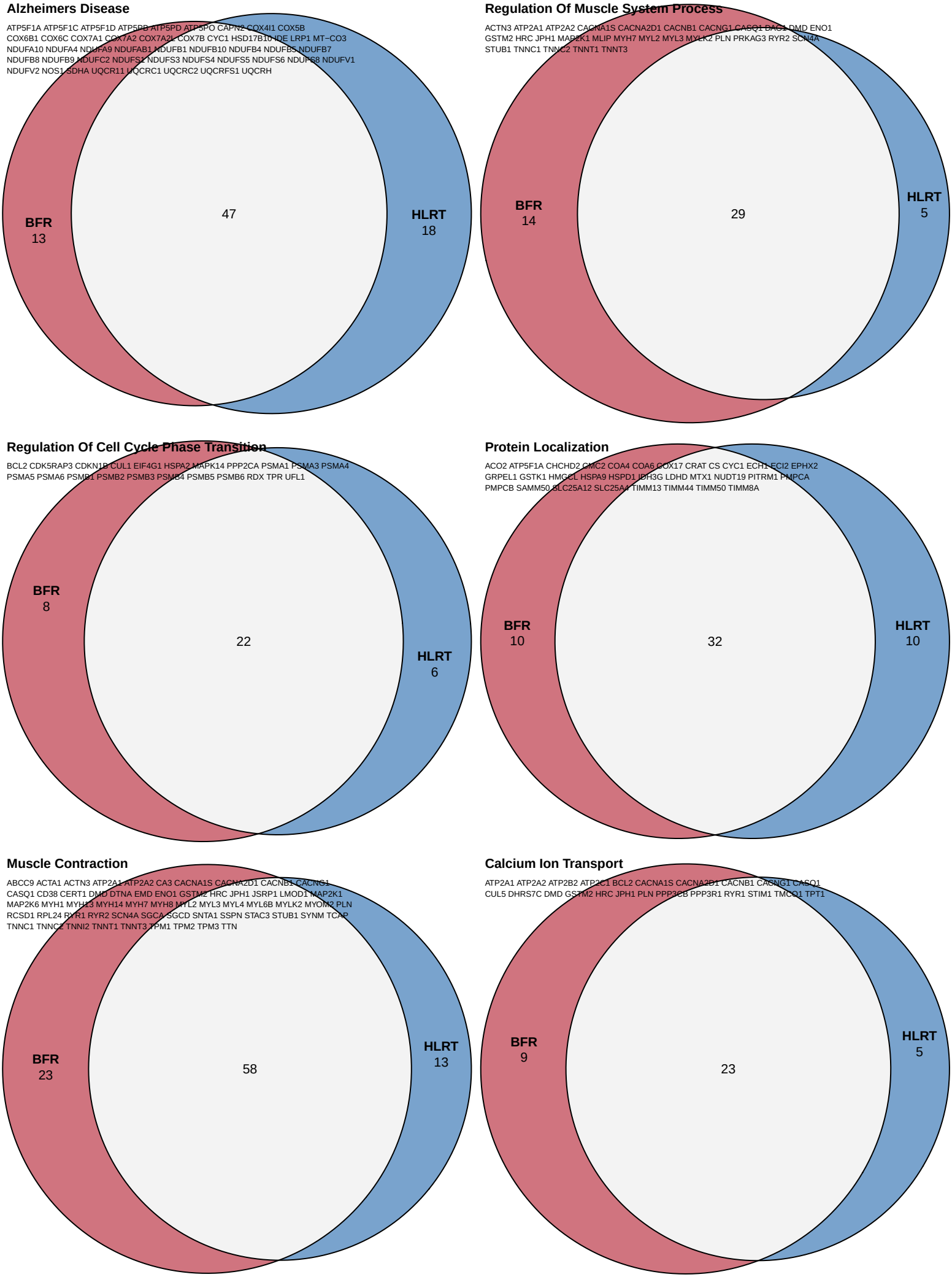
ATP5F1A ATP5F1C ATP5F1D ATP5IF1 ATP5MG ATP5PB ATP5PD ATP5PO CHCHD2 COX4I1 COX5B COX6B1 COX6C COX7A1 COX7A2 COX7A2L COX7B CYC1 GHITM ISCU MT−CO3 MTCH2 NDUFA10 NDUFA12 NDUFA4 NDUFA9 NDUFAB1 NDUFB1 NDUFB10 NDUFB11 NDUFB4 NDUFB5 NDUFB7 NDUFB8 NDUFB9 NDUFC2 NDUFS1 NDUFS3 NDUFS4 NDUFS5 NDUFS6 NDUFS8 NDUFV1 NDUFV2 PPIF SDHA SHMT2 SIRT3 STOML2 UQCR11 UQCRC1 UQCRC2 UQCRFS1 UQCRH



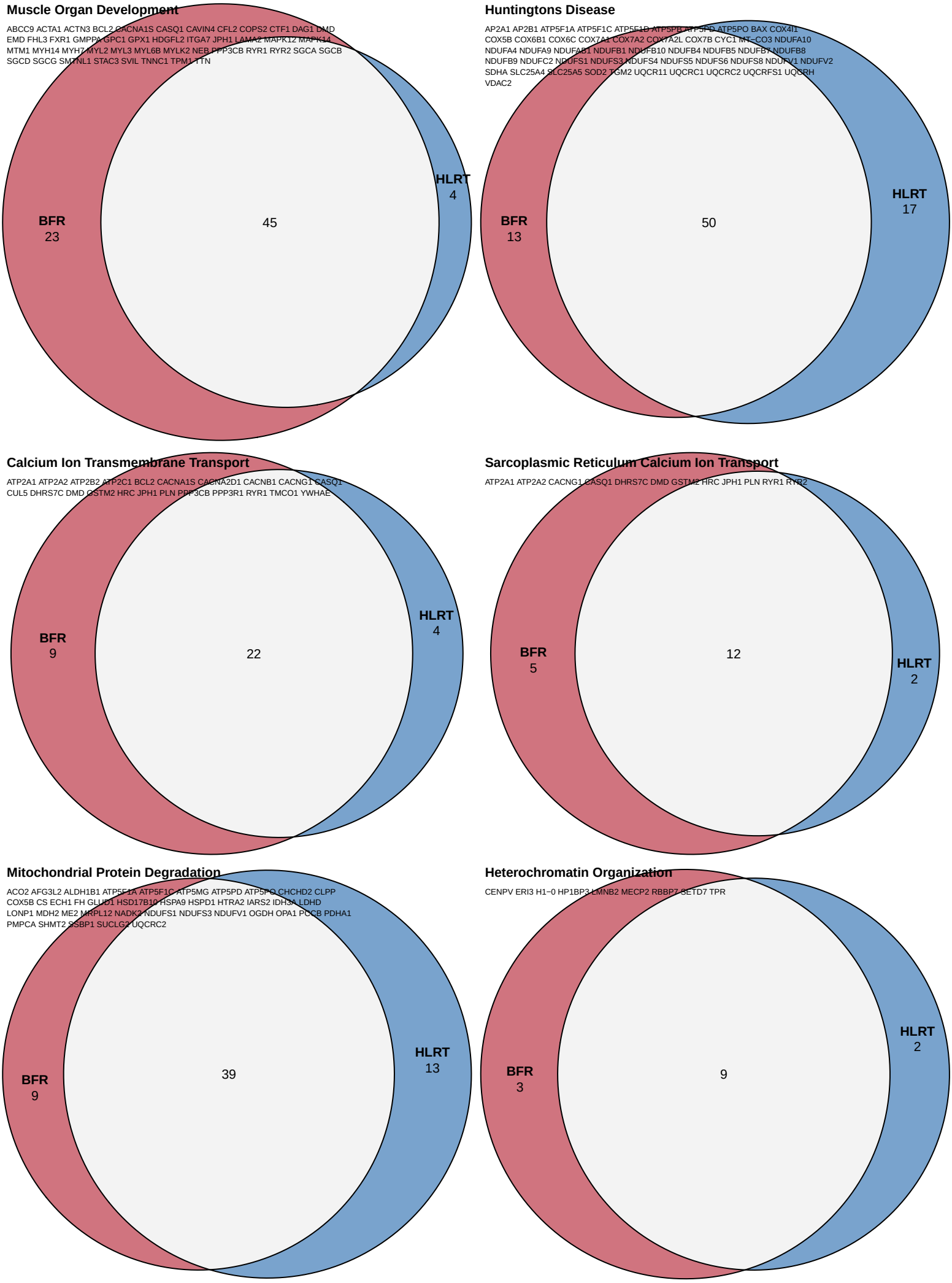
S21 Figure (page 6 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



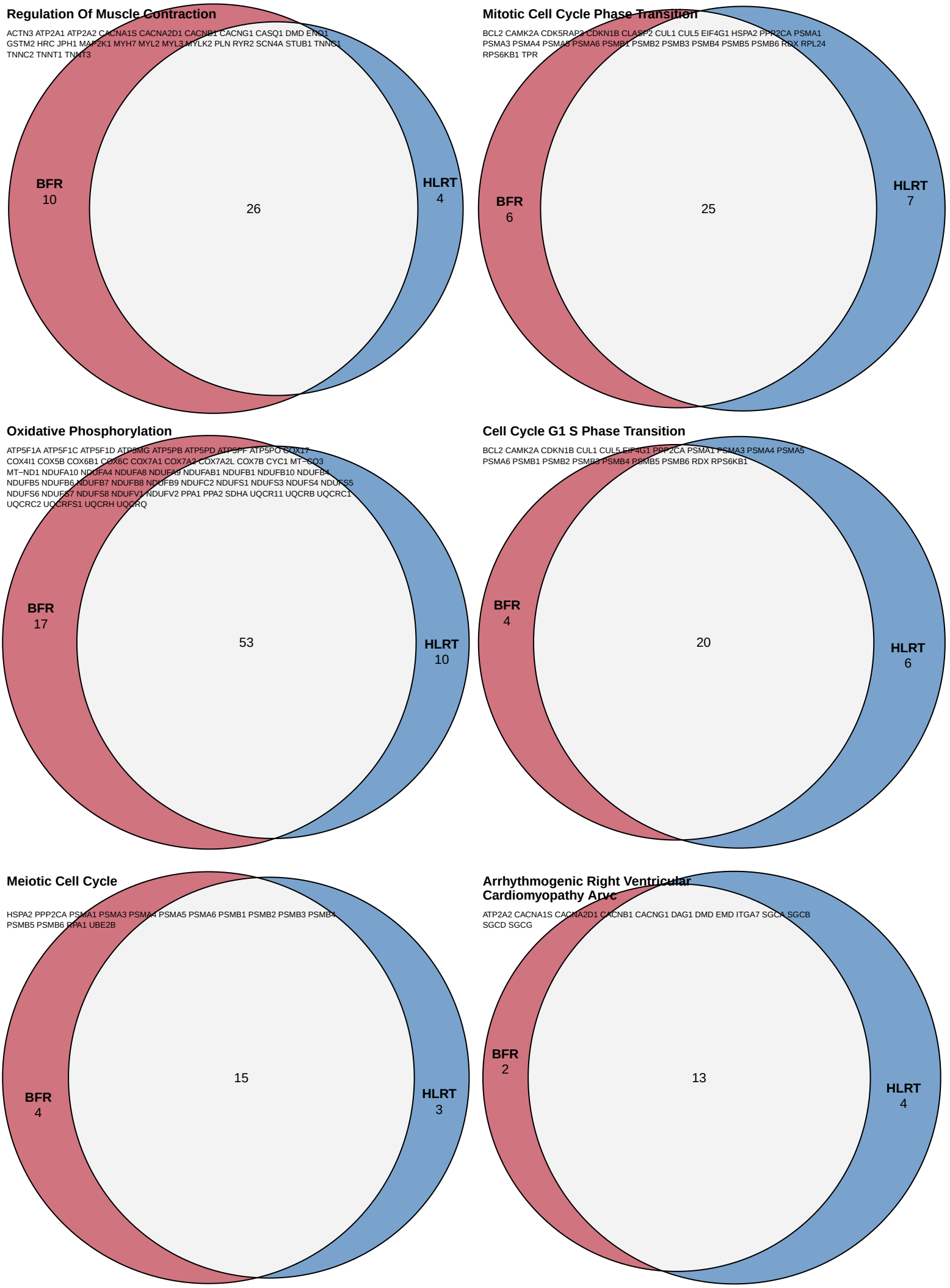
S21 Figure (page 7 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



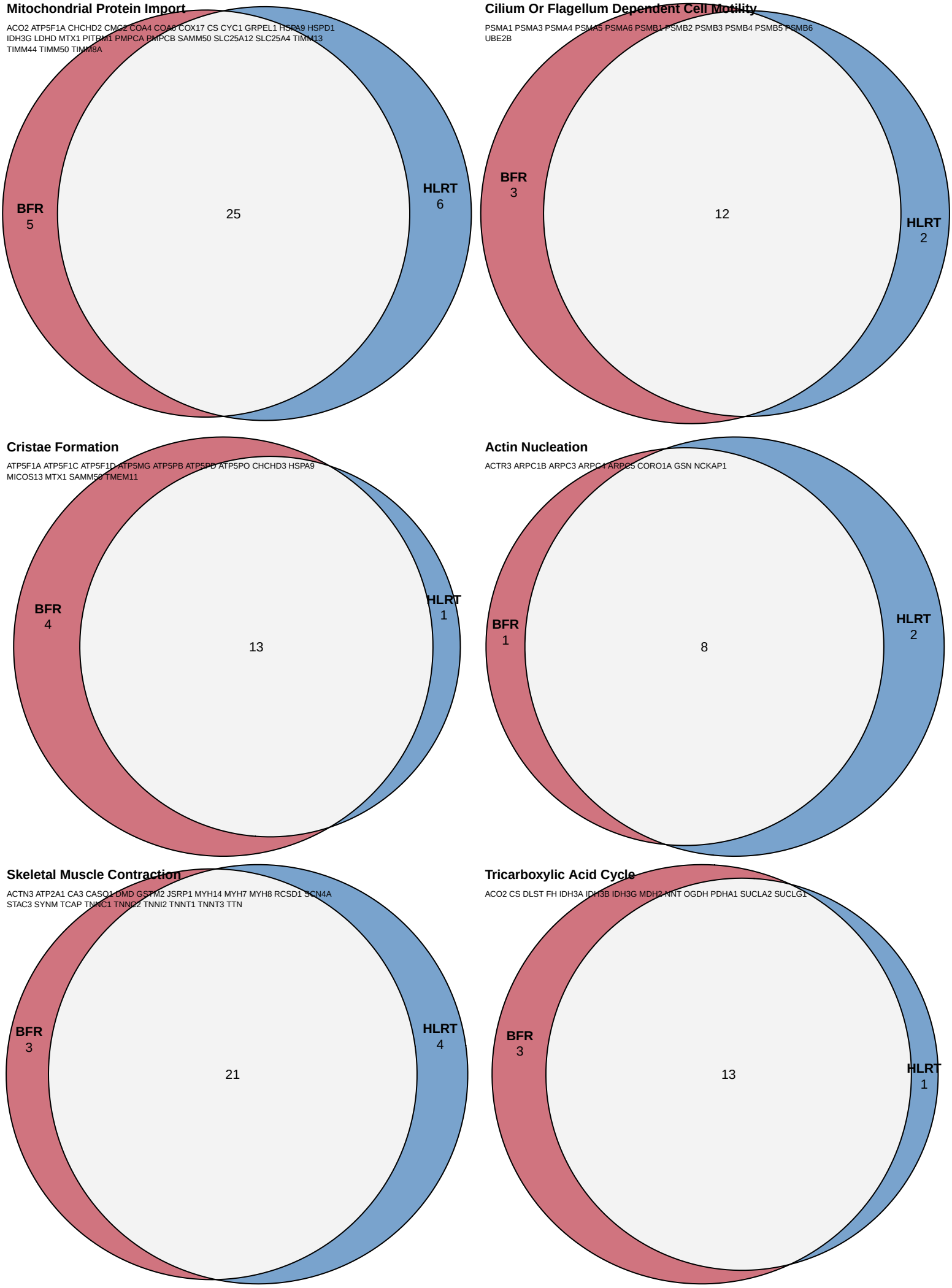
S21 Figure (page 8 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



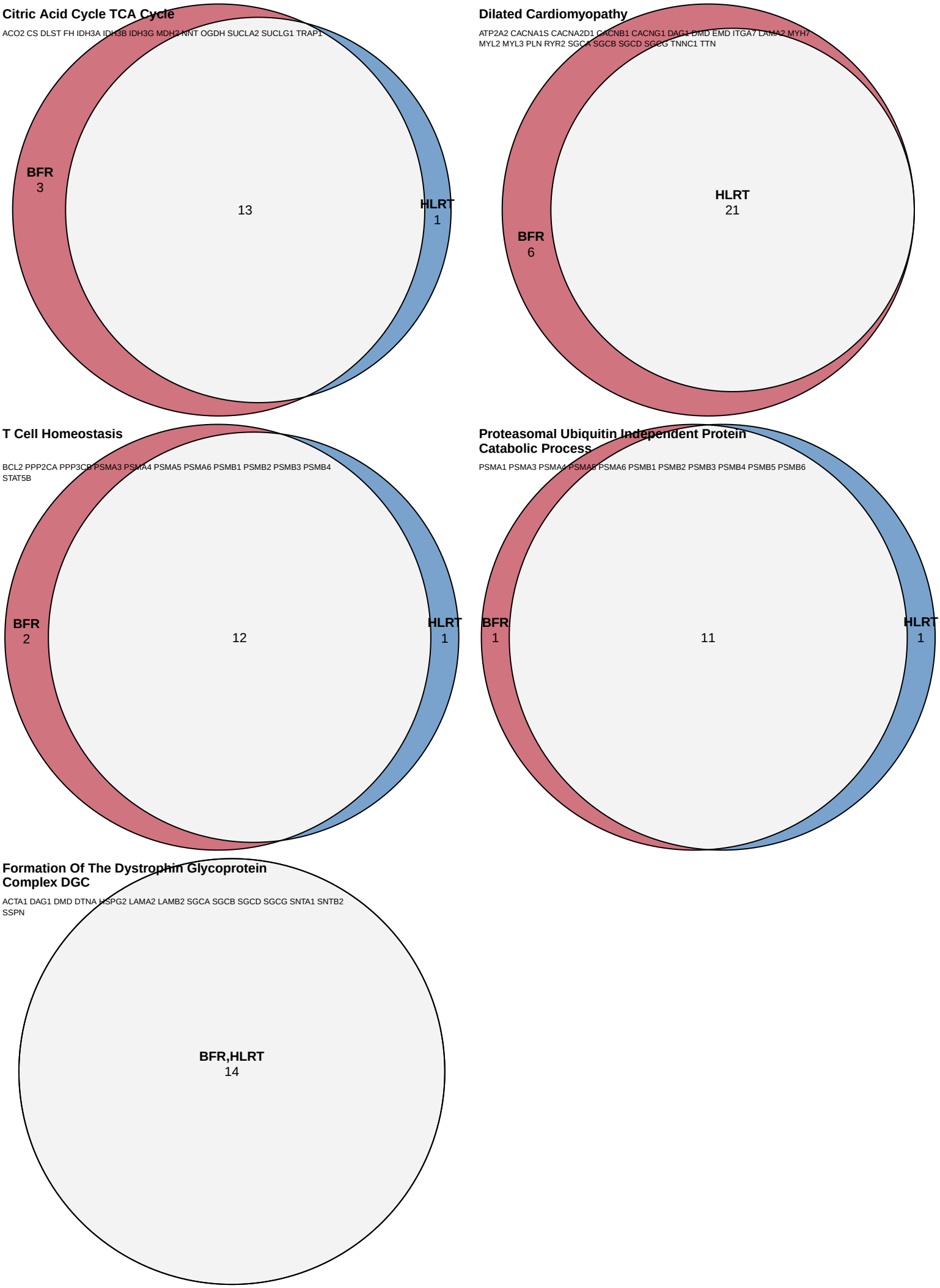
S21 Figure (page 9 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



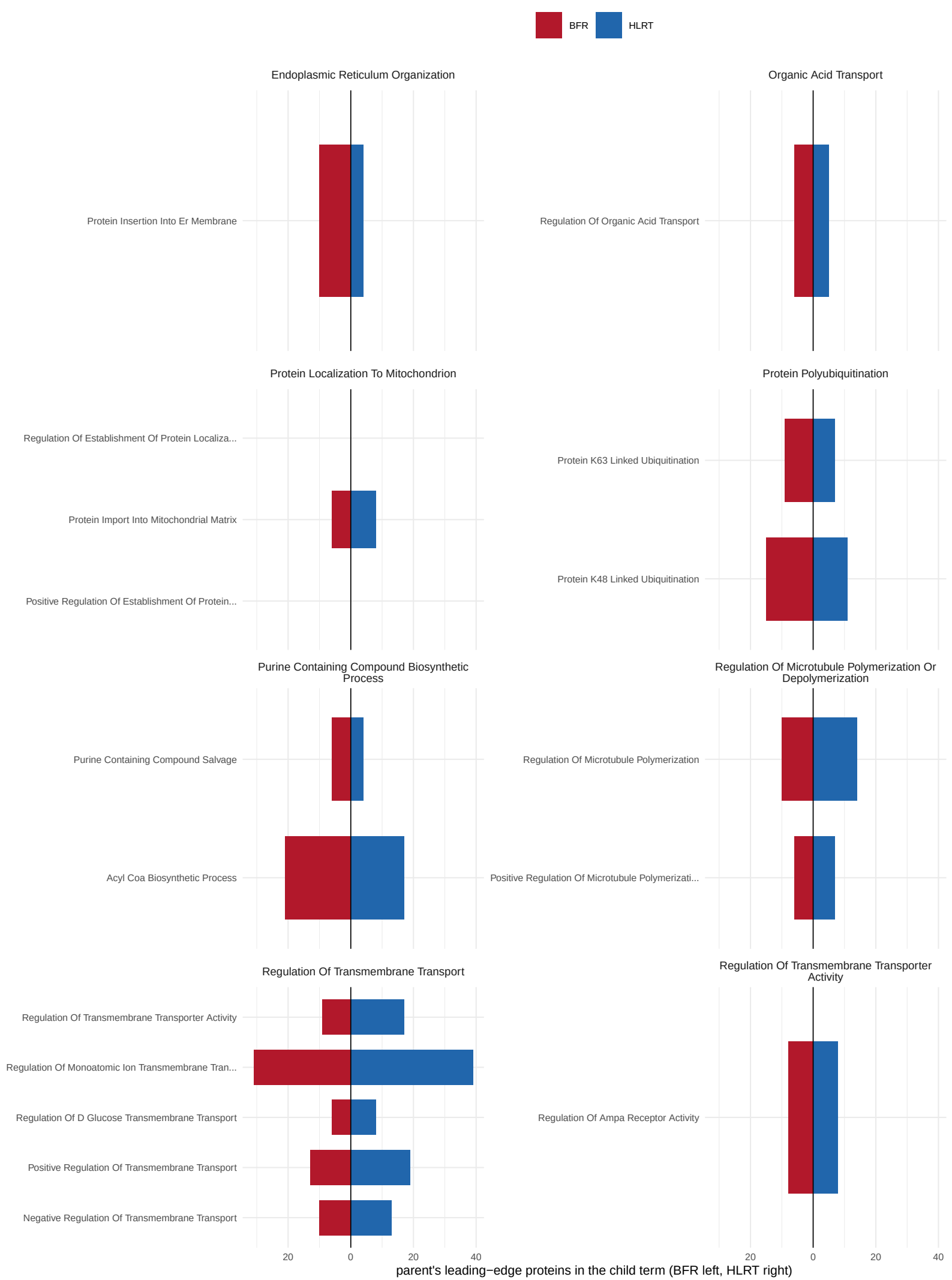
S21 Figure (page 10 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



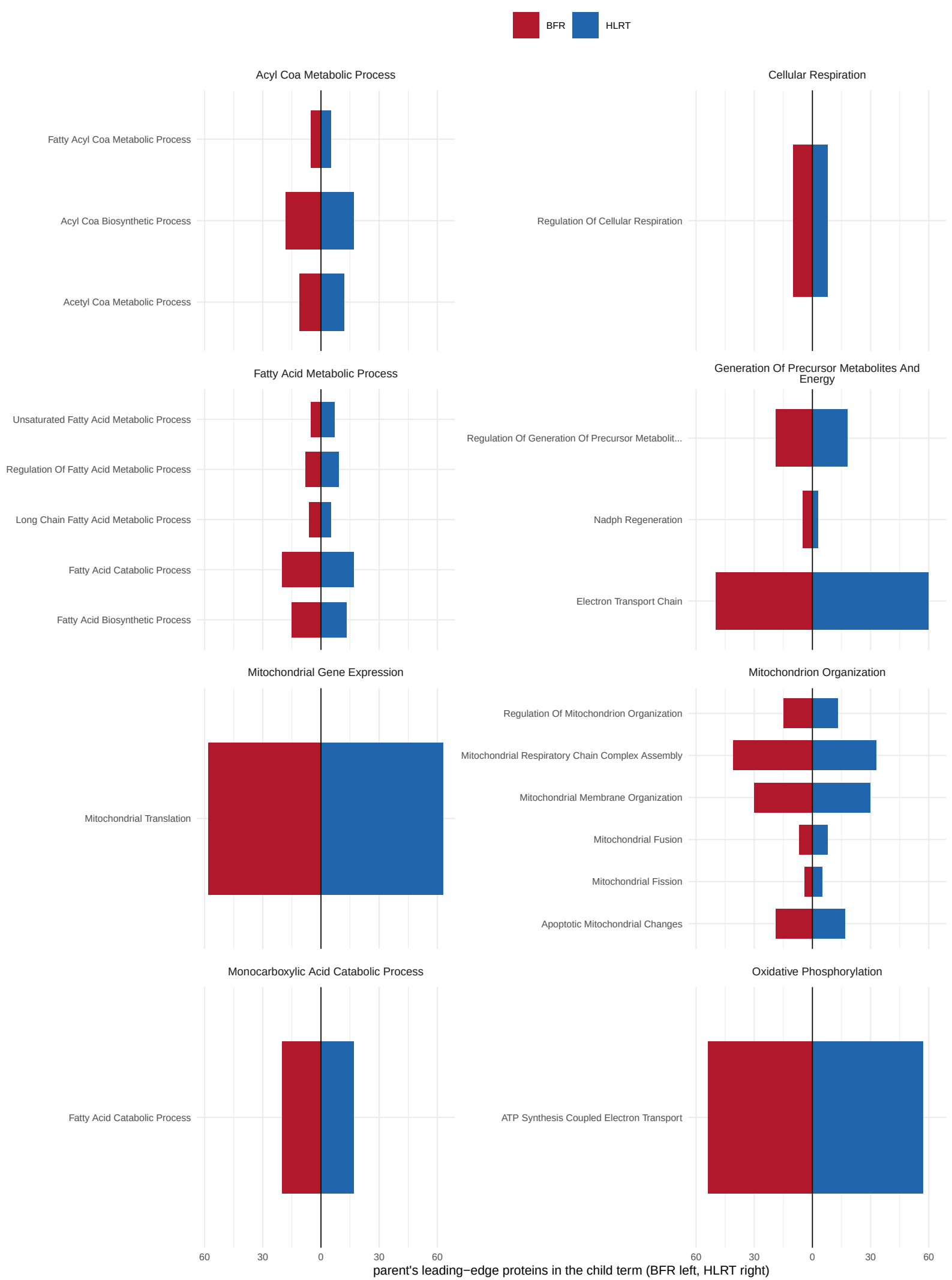
S21 Figure (page 11 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



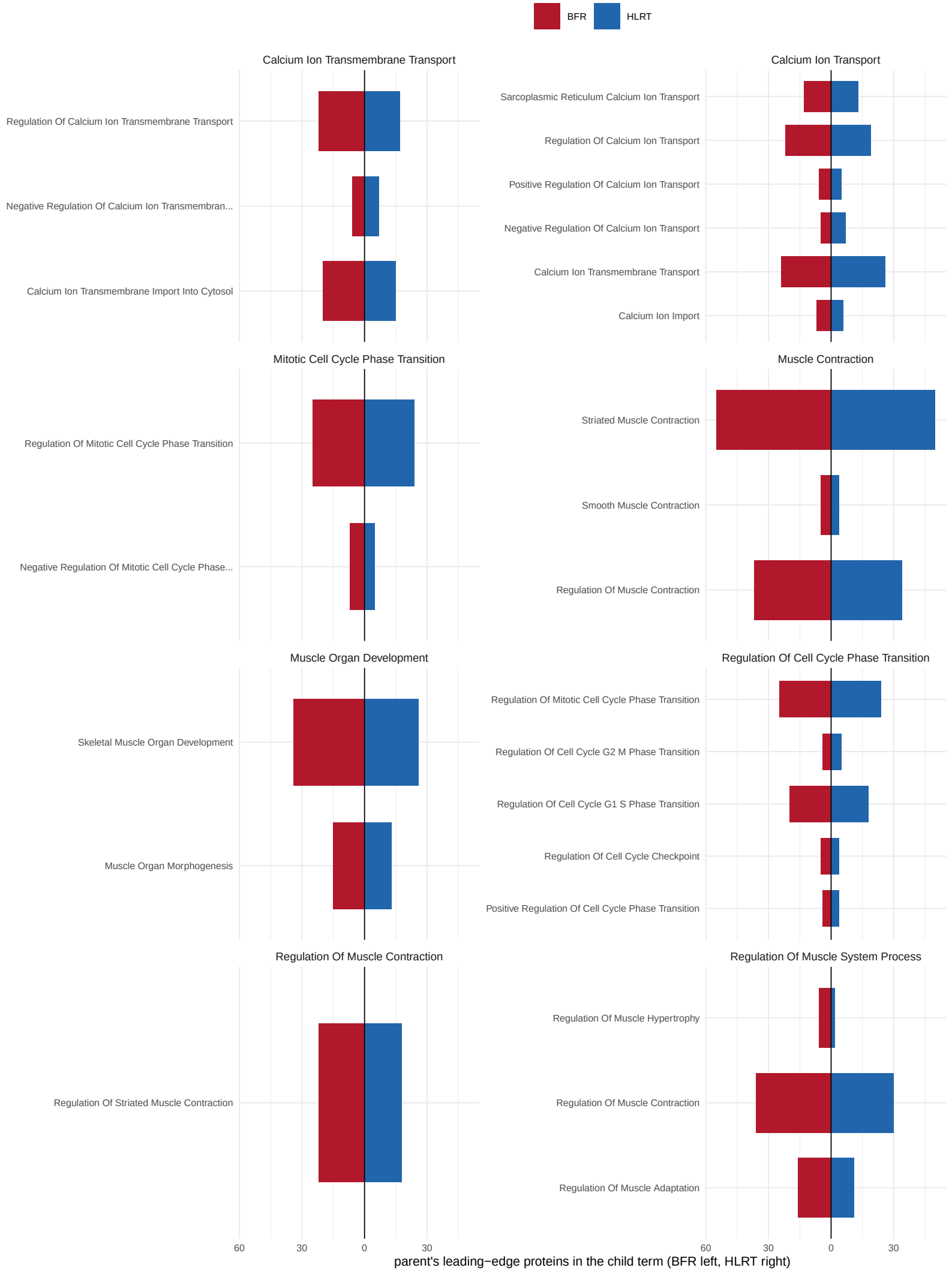
S21 Figure (page 12 of 12). Leading edges by leg, one Venn per pathway in the order of S20. Areas follow the counts of proteins in each leg's leading edge only and in both; the proteins shared by both legs are listed under each title. Data: proteins sheet of 07_leading_edge.xlsx.



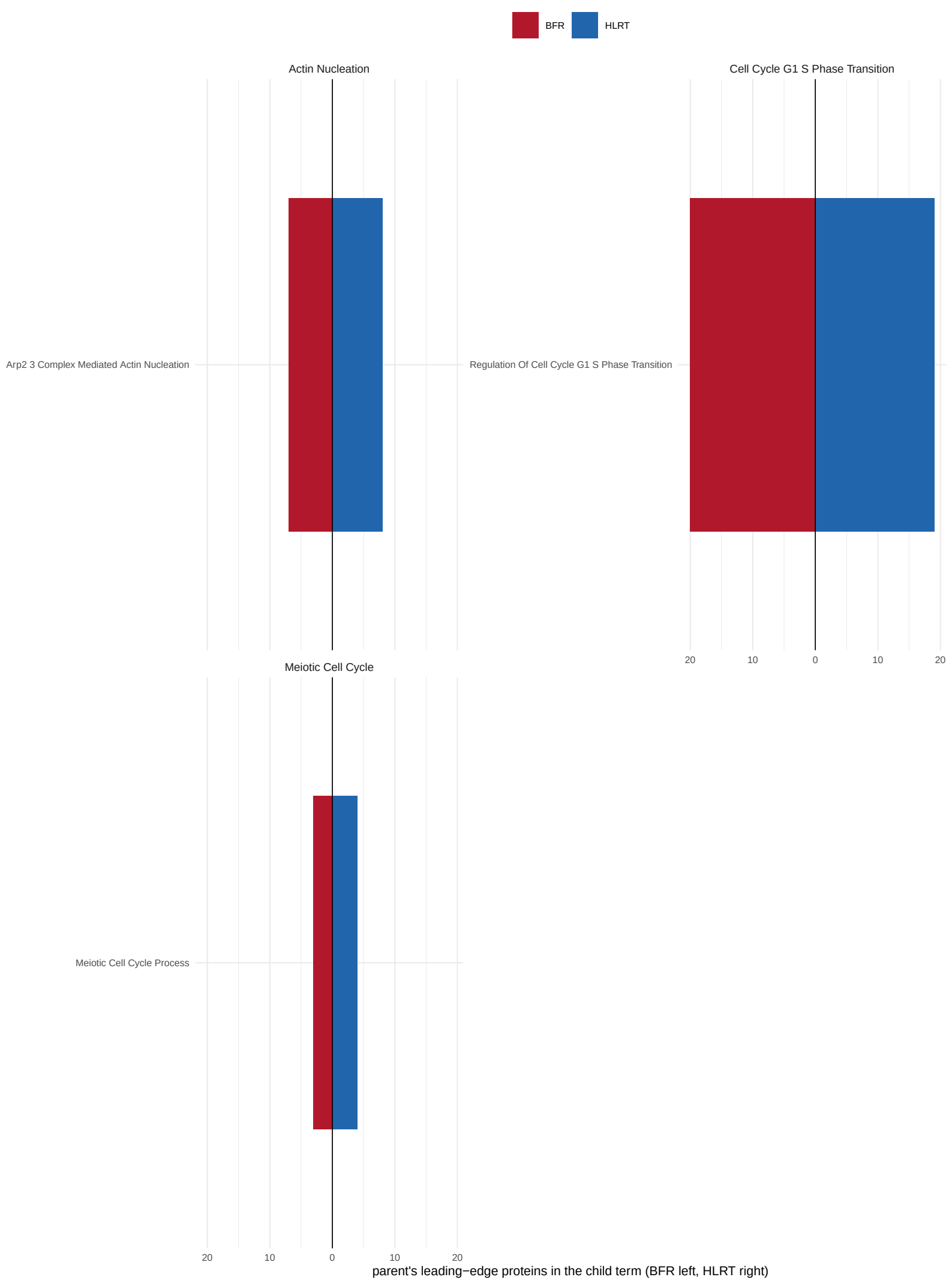
S22 Figure (page 1 of 4). GO:BP pathways both legs move, broken into their tested child terms. Bars count the parent's leading−edge proteins in each child, BFR to the left and HLRT to the right. Child terms overlap, so the counts are not tested. Data: go_children sheet of 07_leading_edge.xlsx.



S22 Figure (page 2 of 4). GO:BP pathways both legs move, broken into their tested child terms. Bars count the parent's leading−edge proteins in each child, BFR to the left and HLRT to the right. Child terms overlap, so the counts are not tested. Data: go_children sheet of 07_leading_edge.xlsx.



S22 Figure (page 3 of 4). GO:BP pathways both legs move, broken into their tested child terms. Bars count the parent's leading−edge proteins in each child, BFR to the left and HLRT to the right. Child terms overlap, so the counts are not tested. Data: go_children sheet of 07_leading_edge.xlsx.



S22 Figure (page 4 of 4). GO:BP pathways both legs move, broken into their tested child terms. Bars count the parent's leading−edge proteins in each child, BFR to the left and HLRT to the right. Child terms overlap, so the counts are not tested. Data: go_children sheet of 07_leading_edge.xlsx.